Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction

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***Abstract*—A conformal prediction region promises to contain a road user's true future path a fixed fraction of the time. We ask whether that promise survives when a trajectory forecaster is calibrated on one driving dataset and deployed, without new labels, on another. Using forecasters calibrated on Argoverse 2 or nuScenes and evaluated on each other and on Waymo Open Motion, we measure coverage losses of 6 to 25 percentage points at a 90 percent nominal level, on two architectures whose accuracy is within 15 percent of published values; the third dataset was registered in advance. Reweighting by an estimated covariate-shift likelihood ratio makes the loss worse, and we show that no label-free method can repair an unrestricted shift. The loss concentrates in turns, most of all in left-hand traffic. Measured input differences between the datasets explain about a quarter of it for one architecture and, with caveats, all of it for the other. Labelled target data can repair it, but labels arrive one scene at a time, and a guarantee that treats agents as independent can fail when scenes contain many agents. A scene-level Learn-then-Test rule with a betting test restores the guarantee at every labelling budget we tested, and returns a region within twice the oracle size from about 150 to 200 labelled scenes. Conformal guarantees for trajectory prediction should therefore not be assumed to transfer across datasets without an audit and a scene-level recalibration.**

***Index Terms*—Conformal prediction, trajectory prediction, distribution shift, autonomous driving.**

AUTONOMOUS vehicles plan around forecasts of what nearby road users will do. A forecast is only as useful as the statement it makes about its own reliability. A planner that knows a predicted region contains the true future with probability at least $1-\alpha$ can keep clear of that region and of nothing larger. Conformal prediction (CP) supplies such a statement for any black-box predictor, with a finite-sample guarantee that needs only exchangeability of calibration and test data [1], [2], and it is now used for trajectory prediction and safe planning [3]–[6].

Deployment breaks exchangeability. A forecaster is calibrated on the data at hand, with one set of cities, one sensor suite and one annotation pipeline, and is then driven somewhere else. Trajectory predictors lose accuracy when moved between datasets [7], [8], and model calibration degrades under dataset shift [9]. What is not known is what happens to the *guarantee itself* when a conformally calibrated forecaster is moved, without new labels, to a different real driving dataset: whether the stated $1-\alpha$ survives, how badly it fails, and whether the standard remedies repair it. Weighted CP needs the likelihood ratio [10], distance-based bounds need the size of the shift [11], [12], and online methods need a stream of labels [13], [14]. None settles the static, label-free, dataset-to-dataset case that Fig. 1 illustrates.

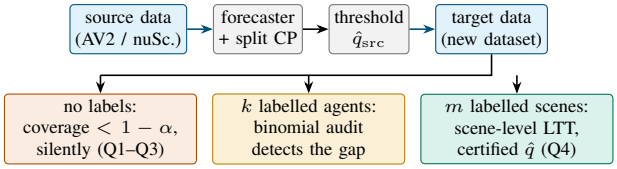


Fig. 1. The transfer problem and the paper's questions. A threshold calibrated on one dataset is deployed on another. Without labels the coverage loss is invisible; k labelled agents detect it; m labelled scenes repair it.

Contributions: (i) A direct measurement of zero-label conformal coverage transfer between real driving datasets, in both directions between Argoverse 2 and nuScenes and, registered before it was run, on Waymo Open Motion, which to our knowledge no prior work has quantified at this scale. Coverage falls by 6.4 to 25.0 points at a nominal 90%, confirmed on two architectures within 15% of published accuracy; within one architecture family the most accurate model loses the most coverage, but this does not hold across architectures. (ii) A diagnosis. Covariate reweighting makes the loss worse, a non-identifiability result (Proposition 1) shows why label-free repair is limited in principle, the loss concentrates in turns in left-hand traffic, and controlled input perturbations account for part of it. (iii) A repair protocol, not a new conformal algorithm: it applies Learn-then-Test with a betting test at the scene level, which gives a finite-sample, training-conditional guarantee when labels arrive scene by scene (Propositions 2 and 3). It is valid at every labelling budget we tested, on both architectures, and gives a usable region from about 150 labelled scenes, while the natural agent-level guarantee can fail when scenes contain many agents and the source threshold is already badly wrong. (iv) Code, registered hypotheses and an append-only experiment log, including a defect found and corrected before submission.

I. Related Work

a) Conformal prediction for trajectories.: Split CP turns any predictor into one with marginal coverage guarantees [1],

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[2] and underlies conformal safe planning around predicted agents [3]. Multi-step and multi-modal trajectories have been handled with copula-based intervals [4], scenario-aware intervals in a Frenet frame on nuPlan with a learned reliability discriminator [15], and multi-view calibration that fuses vehicle- and infrastructure-side observations before conformal calibration [16]. A 2025 survey of the field catalogues these directions and the open gaps around uncertainty and generalisation [17]; concurrent architecture work targets multimodal mode decoding [18] and a shared encoder across simulation, prediction and planning [19], and a controlled dataset-design study on a new benchmark finds that feature engineering matters less than which data the model sees [8] — a finding our Sec. V-C echoes from the calibration side. Calibrating the predictive *distribution* itself, rather than only the output region, has also been proposed [20]. Locally weighted scores that divide the residual by an estimated spread [21] are the classical route to adaptive regions; our normalised score uses the forecaster's own predicted scale in that role, an LLM-based safe-navigation system uses a related idea in token space [22].

b) Safe planning and robot safety with conformal guarantees.: Beyond trajectory regions, conformal prediction has been used to certify learned perception for navigation [23], to adapt a crowd-navigation policy's safety margin to distribution shift [24], to learn safety regions from sparse human feedback [25], to certify learned dynamical-system motion primitives [26], and more broadly to connect conformal prediction to formal verification and control of learning-enabled autonomous systems [6]. Closest to our label-free-versus-labelled distinction, [27] transfers a safety guarantee across policy updates analytically rather than by relabelling, and [28] treats the distribution shift the ego agent itself induces in interacting agents. All of these assume the calibration and deployment *domain* match; we study what happens when the domain itself changes.

c) Conformal prediction under shift.: Weighted CP handles covariate shift given the likelihood ratio [10], and the coverage loss under general shift can be bounded by a distance between calibration and test distributions [11]. Recent work replaces that bound with an explicit ambiguity-set radius under Lévy–Prokhorov perturbations [12], regularises the representation with a Wasserstein penalty to shrink the achievable coverage gap [29], or uses optimal transport with unlabelled target data to both estimate and mitigate the loss in coverage [30]. Robust conformal prediction instead guards against every test distribution inside an f-divergence ball around the training distribution [31]; it restricts the shift class by construction, as Proposition efprop:ident says a useful label-free method must, but needs a radius chosen or estimated for the shift, and we leave it as a baseline for future work. The last is the closest label-free competitor, but its bracketing of the unlabelled test scores is built from the minimum and maximum score over all candidate labels, for scores bounded in $[0, 1]$, and is evaluated on classification and a one-dimensional binned regression task. For our unbounded, metre-scale score over multi-modal trajectories that minimum is zero and the maximum is unbounded, and we found no published construction for structured outputs, so we do not run it as a baseline; a trajectory-specific bracketing is a natural extension. A parallel line gives sample-complexity lower bounds for when a selective predictor can certify a risk floor under covariate shift at all [32], a negative result in the same spirit as our Proposition 1. Online methods adapt the threshold from realised errors: adaptive conformal inference [13], conformal PID control [14], its use for trajectories in CUQDS [5], and adaptive nonconformity scores with a replay buffer [33]. More recent online variants retroactively revise past predictions for faster adaptation [34], cast the threshold update as a universal-portfolio game [35], or give training-conditional (not just time-averaged) regret bounds under drift [36]. The closest trajectory-specific instance of this family is CUQDS [5], which pairs a learned Gaussian-process uncertainty module with online conformal PID control; like the other online methods above, it needs a running stream of ground-truth labels at deployment time, so it answers a different question from ours and is not a baseline we can run in the zero-label or fixed-labelled-batch settings this paper studies. Our ACI comparison (Sec. V-E) covers the same methodological category — adapt the threshold from realised errors — without requiring CUQDS's architecture-specific uncertainty module. Other work conditions on an observable nuisance parameter and generates synthetic data with a diffusion model [37] or treats the shift the ego vehicle itself induces in other agents [28] (both preprints). We study a different point in this space: no feedback, no generative model, and three real driving datasets.

d) Grouped, hierarchical and conditional coverage.: When observations come in groups, exchangeability across observations fails, and hierarchical conformal methods restore validity by pooling or subsampling at the group level [38], [39]; recent extensions handle hierarchically structured outputs with a reconciliation step [40], weight groups directly when the shift is known to act at the group level [41], carry constrained-complexity representations into hierarchical classification [42], or let a test group borrow structure from a donor group when only partial test-group data is available [43]. A broader conditional-coverage literature asks for validity on subpopulations or individual test points rather than only on average: a unifying decomposition of conditional miscoverage [44], learned reliability estimators that predict where coverage will fail [45], [46], group-conditional coverage over overlapping, fractional group memberships [47], an audit showing nominal marginal coverage can still silently fail a specific clinical subgroup [48], and conditional risk control for segmentation via a weighted-quantile connection to risk control [49]. Risk-controlling procedures [50] and Learn-then-Test [51] give finite-sample control of a monotone risk by testing a fixed ordered set of thresholds, and betting-based p-values [52] make such tests far tighter than Hoeffding-type bounds when losses are small. We combine the two ideas for a setting those works do not address: dataset shift in trajectory prediction, where the natural labelling unit is the *scene* and the risk to be controlled is the scene-averaged miscoverage of a fixed, source-calibrated forecaster. Valid p-values for bounded losses need care [53]; our own implementation needed a corrected bet cap (Sec. III-E).

e) Risk control and betting, more generally.: Beyond Learn-then-Test, risk control has been extended to adaptive, data-dependent tradeoffs among several risks at once [54], to a semi-supervised setting that uses unlabelled data to tune the risk-control hyperparameter without extra labels [55], and to post-processing an existing decision policy only where a risk budget is violated, rather than replacing it outright [56]. E-values have been proposed as a more flexible alternative to the p-values we use, enabling anytime-valid and fixed-size variants [57]; weighted-conformal test martingales extend betting-style monitoring to post-deployment shift detection that distinguishes benign covariate shift from harmful shift [58]; and [59] revisits the randomness-theoretic foundations that justify treating a conformal predictor as valid for i.i.d., not only exchangeable, data. We use the same betting machinery for a different purpose: certifying a *scene-averaged* risk from whole labelled scenes.

f) Cross-dataset trajectory prediction and perception under domain shift.: UniTraj harmonises several driving datasets and reports that accuracy drops when models are transferred between them [7]; that study is about point accuracy, not about the validity of an uncertainty guarantee. Beyond trajectories, [9] showed that calibration of deep models degrades under dataset shift; our question is the conformal analogue for structured, multi-modal forecasts. The same cross-dataset gap is documented on the perception side of the autonomous-driving stack: visual localisation uncertainty is miscalibrated under novel weather and lighting [60], collaborative-perception benchmarks struggle to generalise across sensor rigs and sites [61], cross-sensor pretraining is used specifically to close a domain gap that would otherwise need target-domain labels [62], [63], and a broader uncertainty-aware perception survey argues for explicit, communicable uncertainty rather than a single calibrated score [64]. None of this literature asks whether a *conformal* guarantee specifically, rather than calibration quality in general, survives the same transfer; that is the question this paper measures, diagnoses and partially repairs.

II. Problem Setting

A. Conformal regions for multi-modal trajectories

A forecaster f maps a scene x to K candidate futures $\hat{y}_1, \dots, \hat{y}_K \in \mathbb{R}^{T\times 2}$ for a target agent whose true future is y. We use the nonconformity score

$$s(x, y) = \min_{k\le K} \max_{t\le T} \|\hat{y}_{k,t}(x) - y_t\|_2, \tag{1}$$

so that $s \le q$ iff y lies inside the union of the K tube-shaped balls of radius q around the candidates. Given n calibration scenes from a *source* dataset, split CP sets $\hat{q}$ = the $\lceil (n+1)(1-\alpha)\rceil$-th smallest score, and the region $C_{\hat{q}}(x) = \{y : s(x, y) \le \hat{q}\}$ has $\Pr[y \in C_{\hat{q}}(x)] \ge 1-\alpha$ when the test scene is exchangeable with the calibration scenes [1], [2].

B. The transfer problem

Let P_S, P_T be the source and target scene–future distributions. The *coverage gap* of a source-calibrated threshold on the target is

$$\Delta = (1-\alpha) - \Pr_{P_T}\big[s(x, y) \le \hat{q}\big], \tag{2}$$

positive when the guarantee fails (under-coverage). We ask (Q1) how large Δ is on real data; (Q2) whether covariate-shift reweighting [10] removes it; (Q3) whether anything label-free can; and (Q4) how many labelled target agents are needed to detect it, and how many labelled target scenes to repair it.

C. What label-free repair can and cannot do

Covariate reweighting is valid when $P_T(y \mid x) = P_S(y \mid x)$. If the conditional differs—annotation pipelines, sensor noise, and agent behaviour all differ across datasets—no reweighting on x repairs the gap. More strongly, without restricting the class of shifts nothing label-free can.

Proposition 1 (Non-identifiability of a label-free threshold)**.** *Let S_0 be a nonconformity score with a continuous distribution on the target, and consider the scale-shift family $S = c\,S_0$, $c \ge 1$, whose members are all indistinguishable from unlabeled target inputs. Let $\hat{q}$ be any threshold computed without target labels, so its law does not depend on c. If $\Pr[cS_0 \le \hat{q}] \ge 1-\alpha$ for every $c \ge 1$, then $\Pr[\hat{q} = \infty] \ge 1-\alpha$.*

Proof. For fixed $\hat{q} < \infty$, $\mathbf{1}\{cS_0 \le \hat{q}\} \to 0$ pointwise as $c \to \infty$. By dominated convergence, $\mathbb{E}\Pr[cS_0 \le \hat{q} \mid \hat{q}] \to \Pr[\hat{q} = \infty]$, and the coverage requirement gives $\Pr[\hat{q} = \infty] \ge 1-\alpha$. □

Remark 1. *The proposition does not say label-free methods are useless; it says any guarantee* for all *scale shifts is vacuous, so a useful label-free method must either restrict the shift class (as our normalised score does, Sec. III-A) or be complemented by a label-based audit (Sec. III-B).*

D. Cost of using target labels

If k labelled target agents are used directly, the resulting empirical coverage is Beta-distributed around $1-\alpha$ [65], with standard deviation about $\sqrt{\alpha(1-\alpha)/k}$, so ≈ 900 labels are needed for ± 2 points at $\alpha = 0.1$. Detecting a shortfall δ with an exact one-sided binomial test at level ℓ and power $1-\beta$ needs

$$k \approx \left(\frac{z_{1-\ell}\sqrt{\alpha(1-\alpha)} + z_{1-\beta}\sqrt{(\alpha+\delta)(1-\alpha-\delta)}}{\delta} \right)^2 . \tag{3}$$

For $\delta = 0.034$, $\alpha = 0.1$, $\ell = 0.05$, $\beta = 0.1$ this gives $k \approx 750$.

III. Methods

A. Model-normalised score (label-free)

The forecaster emits, per mode k, a Laplace scale $(b^x_{k,t}, b^y_{k,t})$. Let $u_k(x) = \frac{1}{T}\sum_t \sqrt{(b^x_{k,t})^2 + (b^y_{k,t})^2}$ and use the score $s'(x, y) = \min_k \max_t \|\hat{y}_{k,t} - y_t\| / u_k(x)$. The region is

the union of balls of radius $\hat{q}'\,u_k(x)$. The premise is that when a shift makes scenes harder, the forecaster's own scale widens, so the normalised score is less shift-sensitive. Its efficiency is measured by region area $\sum_k \pi(\hat{q}'u_k)^2$ at matched coverage.

B. Coverage audit

Given a source-calibrated $\hat{q}$ and k labelled target agents, an exact one-sided binomial test of H_0: coverage $\geq 1-\alpha$ flags a violation. We report its empirical power and its false-alarm rate on a shift-free split, alongside the theoretical budget of Eq. (3).

C. Recalibration with k target labels

We compare three thresholds computed from k labelled target agents and the n source calibration scores: *direct* (SCP on the k target scores); *pooled* (SCP on the union); and *shrinkage* ($\hat{q} = \lambda\hat{q}_{\text{tgt}} + (1-\lambda)\hat{q}_{\text{src}}$ with $\lambda = k/(k+k_0)$). We report the label budget $k^\star$ at which absolute coverage error falls below 0.02 in at least 90% of random label draws.

D. Certify-or-recalibrate with exchangeable labels

Plain recalibration on k target labels is only *marginally* valid: the coverage of the resulting threshold is $\text{Beta}(\ell, k+1-\ell)$-distributed [65], so it falls below $1-\alpha$ in roughly half of all label draws (Sec. V). A practitioner needs the stronger, *training-conditional* statement [65], [66]: coverage $\geq 1-\alpha$ with probability $\geq 1-\delta$ over the draw of the labels. *Certify-or-recalibrate* (C-or-R) delivers it while keeping the source threshold when it can be certified. With k labelled scores $s_1, \ldots, s_k$ and $M = \#\{s_i > \hat{q}_{\text{src}}\}$:

1) *Certify.* Keep $\hat{q}_{\text{src}}$ iff the one-sided Clopper–Pearson upper confidence bound [67] on its miss probability at level $\delta/2$ is $\leq \alpha$.
2) *Otherwise recalibrate.* Return the ℓ-th smallest labelled score, with ℓ the smallest integer such that $\Pr[\text{Bin}(k, 1-\alpha) \leq \ell - 1] \geq 1 - \delta/2$ (the ℓ-th order statistic has coverage $\text{Beta}(\ell, k+1-\ell)$); if no such $\ell \leq k$ exists return $+\infty$.

Proposition 2 (C-or-R is training-conditionally valid)**.** *If the k target scores are exchangeable with the target scores at test time and continuously distributed, the returned threshold $\hat{q}$ satisfies* $\Pr\left[\Pr_{\textit{test}}(s \leq \hat{q} \mid \textit{labels}) \geq 1-\alpha\right] \geq 1-\delta$.

Proof. Let F be the target score CDF. The event "$\hat{q}$ fails" is contained in (a) the source threshold is certified although $F(\hat{q}_{\text{src}}) < 1-\alpha$, or (b) the branch-2 order statistic has coverage $< 1-\alpha$. (a) has probability $\leq \delta/2$ because the Clopper–Pearson bound is a valid one-sided confidence bound for the miss probability $1 - F(\hat{q}_{\text{src}})$ of the fixed threshold; (b) has probability $\leq \delta/2$ by the definition of ℓ. A union bound gives δ. □

Certification needs slack: a source threshold calibrated at exactly α has miss probability $\approx \alpha$ and its confidence bound cannot fall below α; in our experiments it is (almost) never certified on shifted data, and on unshifted data it is certified only if calibrated at a smaller $\alpha_{\text{src}} < \alpha$.

E. Scene-level risk control under clustered labels

Driving datasets label many agents per scene, and agents in one scene share road, traffic and annotator. Labels drawn scene by scene are not exchangeable at the agent level, and Proposition 2 then fails (Sec. V). We therefore make the *scene* the unit. For a labelled scene j with agent scores $\{s_{ji}\}$ let $L_j(q) = \frac{1}{n_j}\sum_i \mathbf{1}[s_{ji} > q] \in [0,1]$, non-increasing in q, and control the scene-averaged miscoverage $R(q) = \mathbb{E}_j\, L_j(q)$, which weights every *scene* equally regardless of how many agents it contains. This is a different quantity from the agent-pooled miss rate $\frac{1}{\sum_j n_j}\sum_{j,i}\mathbf{1}[s_{ji} > q]$, which weights every agent equally and so is dominated by whichever scenes happen to be largest; the two coincide only when every scene has the same agent count. We control $R(q)$ because it is the quantity the scene-level test (below) can be made valid for under scene-level exchangeability, and because it is the natural risk when scenes, not agents, are the unit a planner reasons about. Given m labelled scenes, scene-level certify-or-recalibrate proceeds as in Sec. III-D but with Learn-then-Test [51] over a *fixed*, label-independent grid $q_1 > q_2 > \cdots > q_G$ (log-spaced around the source threshold, $\hat{q}_{\text{src}} \times [4, 0.5]$, $G = 300$): for each candidate we compute a p-value for $H_q : R(q) > \alpha$ from the m per-scene losses, test candidates from the largest downwards at level $\delta/2$, and return the last q rejected ($+\infty$ if none). The source threshold is tried first at level $\delta/2$ (certification). Two p-values are used: Hoeffding–Bentkus [50], and a betting p-value [52]: with predictable, variance-adaptive bets $\lambda_i \in [0, \frac{1}{2(1-\alpha)}]$ the capital $K = \prod_i \big(1 + \lambda_i(\alpha - L_i)\big)$ stays non-negative for $L_i \in [0,1]$ and is a supermartingale under $\mathbb{E}L \geq \alpha$, so $p = \min(1, 1/\max_t K_t)$ is valid by Ville's inequality. The cap matters: any bet above $1/(1-\alpha)$ lets a single fully missed scene drive the capital negative, which voids the argument.

Proposition 3 (Scene-level guarantee)**.** *If the m labelled scenes are exchangeable draws from the target scene population, and each p-value is valid for its null, the returned threshold satisfies* $\Pr[R(\hat{q}) \leq \alpha] \geq 1-\delta$.

Proof. R is non-increasing, so along the descending grid the false nulls ($R(q_i) \leq \alpha$) form a prefix and the true nulls a suffix beginning at some i_0. The procedure returns a threshold with $R > \alpha$ only if it rejects $H_{q_{i_0}}$ (fixed-sequence testing stops at the first non-rejection), which has probability $\leq \delta/2$ because the grid does not depend on the labels; the certification branch fails with probability $\leq \delta/2$. A union bound gives δ. □

F. Online baseline

Adaptive conformal inference (ACI) [13] updates the working level $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ after every revealed label, keeping the source calibration set fixed; it needs a stream of labels and offers long-run, not finite-sample, coverage. We report (i) how many streamed labels ACI needs before its running coverage stays within 2 points of nominal, and (ii) the coverage of its threshold frozen after k labels, for $\gamma \in \{0.005, 0.01, 0.05\}$.

G. Controlled input shifts

To locate the source of the loss we perturb one measured AV2–nuScenes difference at a time in AV2 test *inputs* (ground truth untouched) and record the coverage lost by the AV2-calibrated threshold: (i) *2 Hz history*: keep the five 2 Hz keyframes of the 2 s history and interpolate linearly; (ii) *lane dropout*: drop each map polyline with the probability implied by the measured median number of lanes near the target agent (nuScenes 52 vs. AV2 85); (iii) both; (iv) *scene mix*: re-weight AV2 scenes to the nuScenes distribution of seven label-free scene factors with cross-fitted domain-classifier weights, which isolates pure covariate shift. Two further candidates were *measured* and excluded: excess history noise (a quadratic-fit residual estimate is smaller on nuScenes: 0.13 m vs. 0.29 m) and map-point density (equal per scene).

H. Label-free shift monitor

A domain classifier on label-free scene features (native sampling interval, nearby-agent count, map density, nearby lanes, history speed, heading change, curvature) is trained to separate source and target scenes; its AUC is the monitor. Within a dataset, the same classifier is trained per city against the rest. We test whether the AUC predicts the coverage loss (across cities, by Spearman correlation).

IV. Experimental Setup

a) Data: Argoverse 2 [68], nuScenes [69] and Waymo Open Motion [70] (30 of 150 validation shards), converted with ScenarioNet [71] and loaded through UniTraj [7]. AV2 scenarios are independent 11 s clips with one focal agent; the labelled AV2 validation split is divided by a fixed salted hash into a calibration half and a test half. nuScenes has a mean of 66 agents per 20 s scene (9041 from 138 validation scenes), so its calibration/test split is made by scene (64/74 scenes). Waymo scenarios carry 3.7 agents on average and are split by scenario (4331 calibration, 4318 test).

b) Forecasters: AutoBot-Ego [72] with $K = 6$ modes. The *confirmatory* models were trained after the hypotheses about them were registered, with UniTraj's recipe (effective batch 128, learning rate 7.5×10^{-4} halved at epochs 10–50): **av2_gpu_full** on the full 183k-scene AV2 training set and **ns_gpu_full** on the full nuScenes training split. Each had to reach validation minADE_6 within 15% of UniTraj's published value [7]; they reached 0.854 (gate 0.98, reference 0.85) and 1.232 (gate 1.39, reference 1.21). The final model is the last checkpoint, fixed in advance. A second architecture, Wayformer [73], was trained forward-only under the same accuracy gate and effective batch size, with its own learning rate and 27-epoch schedule (reaching 0.932, Sec. V-F); no reverse-direction Wayformer was trained. Four *reduced-budget* AutoBot models (training-data subsets or shorter schedules; the three AV2 models reach minADE_6 1.09–1.35) were used to design the analyses and are labelled exploratory throughout.

c) Protocol: $\alpha = \delta = 0.1$. Forward: calibrate on the AV2 calibration half; test on the AV2 test half (same-domain control), nuScenes and Waymo. Reverse: calibrate on the nuScenes calibration scenes; test on nuScenes test scenes, AV2 and Waymo. Label-budget experiments draw k agents or m whole scenes 1000 times from a labelled pool that is disjoint from the evaluation pool. Intervals on multi-agent targets resample scenes.

d) Registration and disclosures: Hypotheses, thresholds and success criteria were written to a version-controlled, timestamped log (released with the code) before the runs they govern; later analyses were added as dated addenda, and outcomes are appended, never edited. Defects found during the work are disclosed there: a cache collision, per-agent instead of per-scene nuScenes splits, and truncated scenario identifiers, none affecting a reported number; and a betting p-value whose bet cap allowed negative capital. That last defect was found in an internal review before submission, a regression test that detects it was added, and every repair result in Sec. V-E was recomputed with the corrected code under a design registered before the rerun.

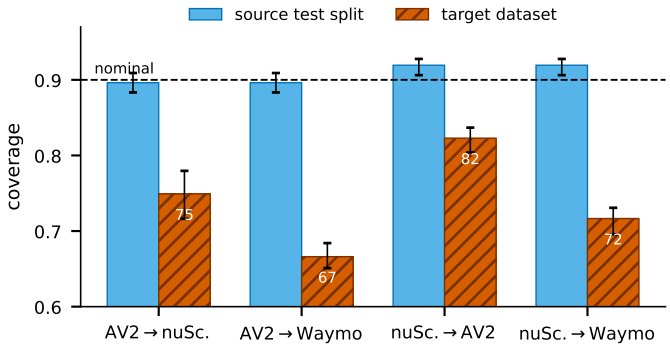


Fig. 2. Coverage of the source-calibrated region ($\alpha = 0.1$) on the source test split and on the target dataset, confirmatory models. Error bars: 95% intervals, resampled by scene where targets have several agents per scene (nuScenes, Waymo).

V. Results

Confirmatory results come first, organised by the four questions of Sec. II. The reduced-budget models on which the analyses were designed are summarised in Sec. V-G.

A. Q1: How large is the gap?

On its own test split each confirmatory model is calibrated as promised: the same-domain gap is 0.4 points for the AV2 model and -1.9 points for the nuScenes model (Fig. 2). On every other dataset it under-covers. Calibrated on AV2, coverage falls by 15.1 points on nuScenes and by 23.4 points on Waymo. Calibrated on nuScenes, it falls by 7.7 points on AV2 and by 18.3 points on Waymo. All four intervals exclude zero by a wide margin, so a Holm correction over the four registered tests changes no conclusion. The Waymo tests were registered before any Waymo prediction was computed. In metres, the AV2-calibrated tube has radius 3.8 m, while 90% coverage needs 8.2 m on nuScenes and 8.9 m on Waymo: the source region is less than half as wide as the target requires.

The nuScenes test agents come from only 138 scenes, so they are strongly dependent. Resampling whole scenes inflates the variance of the coverage estimate by a design effect of 3.9;

TABLE I
COVERAGE GAP BY MODEL ($\alpha = 0.10$, POINTS; TARGET NUSCENES FOR FWD., AV2 FOR REV.). CONF.: GATED FULL-DATA MODEL; EXPL.: REDUCED-BUDGET MODEL USED TO DESIGN THE ANALYSES. NORM.: GAP OF THE NORMALISED SCORE. REW.: CHANGE IN GAP FROM COVARIATE REWEIGHTING (POSITIVE = WORSE). AUDIT: REJECT PROBABILITY / FALSE-ALARM % AT $k = 1000$ LABELS.

Model	St.	Dir.	gap	95% CI	norm.	rew.	audit
av2_gpu_full	conf.	fwd	15.1	13.6–16.7	11.0	+1.9	1.00/7
ns_gpu_full	conf.	rev	7.7	6.3–9.6	8.6	+0.1	1.00/0
av2_cpu_v1	expl.	fwd	3.3	2.1–4.5	1.7	+2.0	0.95/2
av2_valsplit_v1	expl.	fwd	8.0	6.4–9.4	7.0	+2.1	1.00/1
av2_cpu_v2	expl.	fwd	10.5	9.4–12.2	7.3	+2.2	1.00/6
ns_cpu_v1	expl.	rev	4.7	3.7–6.2	7.1	-0.5	1.00/19

all intervals on multi-agent targets therefore resample scenes, not agents. The more accurate model has the larger gap. The confirmatory AV2 model, within 0.5% of UniTraj's published accuracy, loses 15.1 points, against 3.3 to 10.5 points for the three reduced-budget AV2 models (Table I). With four models of one architecture family this is an association, not a causal claim.

B. Q2: Does covariate reweighting fix it?

No. Weighting the calibration scores by an estimated, label-free likelihood ratio increased the gap on every forward model by about two points (column "rew." in Table I) and left the reverse direction unchanged. Reweighting AV2 scenes to the nuScenes mix of speeds and manoeuvres even moves coverage the other way (Table II): nuScenes' scene mix is easier, so composition cannot explain the loss. This is the pattern expected when the conditional distribution of the future given the scene changes, which no reweighting on the scene can repair (Proposition 1).

C. Where does the loss come from?

The loss is concentrated in turns. On the confirmatory model, right-turn coverage falls by 41 points from AV2 to nuScenes, against 11.2 points for straight driving (Fig. 3). It is largest in Singapore, where traffic drives on the left: the right-turn drop there exceeds the drop in Boston by 27 points. What AV2-trained forecasters miss is the wide, across-traffic turn of left-hand traffic. For AutoBot, injecting one measured input difference at a time into AV2 test inputs reproduces only part of the gap (Table II): a 2 Hz history explains 15%, lane dropout 12%, and both together about a quarter. Excess annotation noise and map-point density were measured and found absent.

D. Q3: What can label-free methods do?

Little, and not reliably. The model-normalised score (Sec. III-A) reduces the AV2→nuScenes gap by 27% (to 11.0 points) but widens the reverse gap (to 8.6 points). On the reduced-budget models its effect ranges from a 49% reduction to a 52% increase. A domain-classifier shift monitor correlates with per-city coverage loss at Spearman $\rho \leq 0.47$ on every model, below the registered bar of 0.5. Adaptive conformal

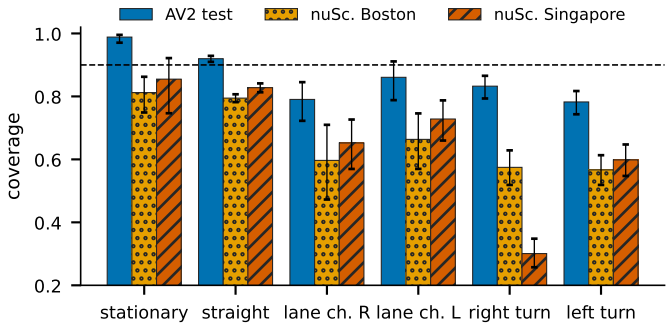


Fig. 3. Coverage by manoeuvre for the confirmatory AV2 model ($\alpha = 0.1$, dashed = nominal). nuScenes is split by city; Singapore drives on the left. Error bars: agent-level Wilson 95% intervals.

TABLE II
WHERE DOES THE LOSS COME FROM? COVERAGE LOST (POINTS; SHARE OF THE REAL GAP G IN PARENTHESES) WHEN ONE MEASURED AV2–NUSCENES DIFFERENCE IS INJECTED INTO AV2 TEST INPUTS: 2 HZ HISTORY, LANE-GRAPH DROPOUT, BOTH, AND RE-WEIGHTING AV2 SCENES TO NUSCENES' SCENE MIX. EXCESS ANNOTATION NOISE AND MAP-POINT DENSITY WERE MEASURED AND FOUND ABSENT (NUSCENES IS SMOOTHER; EQUAL DENSITY), SO THEY ARE NOT INJECTED.

Model	Set	G (pt)	2 Hz	lanes	both	scene mix
av2_cpu_v1	expl.	+3.3	+0.6 (19%)	+1.4 (43%)	+1.8 (53%)	-1.3 (-38%)
av2_valsplit_v1	expl.	+8.0	+0.7 (8%)	+1.4 (17%)	+1.4 (17%)	-1.4 (-18%)
av2_cpu_v2	expl.	+10.5	+1.1 (11%)	+1.3 (12%)	+2.2 (21%)	-1.1 (-11%)
av2_gpu_full	conf.	+15.1	+2.3 (15%)	+1.8 (12%)	+4.1 (27%)	-1.0 (-6%)

inference [13] converges only with a stream of labels; frozen after 1000 labels, its threshold covers $\geq$ 90% of held-out scenes in only 40–50% of streams. These results agree with Proposition 1: without target labels there is no general fix.

One label-free method restricts the shift class, as that proposition says a useful one must. *Protocol-matched recalibration* passes the source calibration inputs through the measured AV2→nuScenes input differences of Sec. III-G (2 Hz history and lane dropout, estimated from target inputs only) and recalibrates on the result. It is exploratory and carries no guarantee. For AutoBot it moves the threshold from 3.8 to 4.7 m and coverage on nuScenes from 75% to 81%, closing 41% of the gap. For Wayformer it moves it from 4.2 to 5.7 m, coverage from 84% to 92%, and overshoots the gap to -2.3 points, in the conservative direction, which may partly reflect how Wayformer's velocity inputs are reconstructed for injected trajectories (Sec. V-F). It does not shorten the label budget of Sec. V-E: started from the matched threshold, the scene-level rule still needs $m = 150$ scenes (AutoBot) and $m = 150$ (Wayformer) for a usable region. A measured, label-free correction can recover part of the gap, but it does not replace labels.

The loss is not an artefact of the registered score. We re-scored the same transfers with four alternatives (exploratory): the top-probability mode only, final displacement, a probability-weighted score, and a score scaled per time step by the source median error. Every one of the 24 resulting out-of-domain cases under-covers (24 of 24), by 4 to 26 points. No score is uniformly better: the time-scaled score cuts AutoBot's AV2→Waymo loss from 23 to 9 points but raises

its AV2→nuScenes loss from 15 to 24.

E. Q4: How many labels to audit and to repair?

Three label budgets answer three different questions: how many labels to *detect* that the guarantee has failed, how many to return a threshold with a *certified* (training-conditional) guarantee, and how many before that threshold is also *small enough to be useful*. We report all three; the second does not imply the third.

Audit (detect). With $k = 1000$ labelled target agents the exact binomial audit (Sec. III-B) detects the gap with probability at least 1.00 on both confirmatory models, at a false-alarm rate of at most 7% on a shift-free split (Table I).

Repair with exchangeable labels (certify). When $k = 1000$ agents are labelled at random, certify-or-recalibrate (Sec. III-D) covers $\geq 90\%$ of held-out agents in 94–98% of label draws, against 50–60% for plain recalibration, at a region 1.18–1.23× the oracle's.

Repair with whole-scene labels (certify at the scene level). Real labels arrive scene by scene, which breaks agent-level exchangeability. We draw m labelled scenes from a pool disjoint from the evaluation scenes (nuScenes train for AV2→nuScenes, the Waymo calibration half for the Waymo targets) and evaluate the scene-averaged miscoverage $R(\hat{q})$ exactly on the evaluation pool, 1000 draws per m (Fig. 4). The scene-level rule with the betting p-value is valid at every m on all four transfers ($\Pr[R(\hat{q}) \leq \alpha] \geq 0.94$). The agent-level rule is not: on AV2→nuScenes, with a mean of 66 agents per scene (median 53), it is valid in 56% of draws at $m = 40$ and 4% at $m = 300$, and one-agent-per-scene split CP [38] in 57%. On Waymo and AV2, with few agents per scene, the agent-level rule stays valid in at least 65% of draws: agent-level guarantees fail where scenes are large.

Useful. Validity is not usefulness. With $m \leq 40$ scenes the scene-level rule abstains in every draw, correctly reporting that it cannot certify any finite threshold rather than returning an unsupported one. The abstention rate falls with m but is not negligible close to the recommended budget: on AV2→nuScenes it is 57% at $m = 80$ and still 6% at $m = 100$. It returns a usable region (at most twice the oracle area, abstaining in $\leq 5\%$ of draws) from $m = 150$ scenes for AV2→nuScenes, AV2→Waymo and nuScenes→Waymo, and from $m = 200$ for nuScenes→AV2, whose single-agent scenes carry less information each. At that budget the repaired AV2→nuScenes radius is 11.3 m against an oracle of 8.2 m. The betting p-value gives smaller regions than Hoeffding–Bentkus at large m (1.6× vs. 2.1× the oracle area at $m = 200$ on AV2→nuScenes). In dataset terms, 150 scenes is more than the 138 scenes of the entire nuScenes validation split, so on that target the repair draws on training-split labels; a deployment that can label fewer scenes should expect the rule to abstain, not to return a smaller guaranteed region.

F. A second architecture: Wayformer

Every result above uses AutoBot. We trained Wayformer [73] on the same full AV2 set under the same gate (val minADE_6 0.932, within 15% of UniTraj's reference, matching AutoBot's gate) to ask whether the findings are AutoBot-specific. No reverse-direction Wayformer was trained.

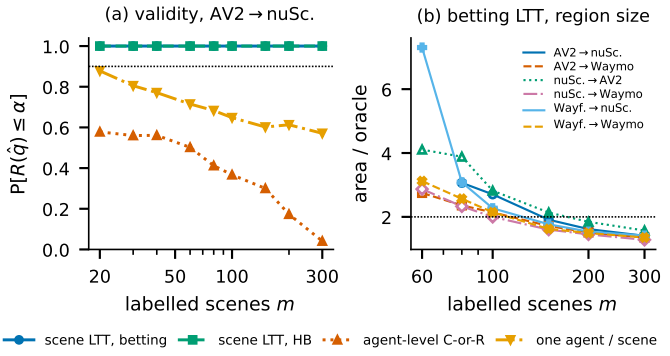


Fig. 4. Repair with m whole labelled scenes, disjoint label and evaluation pools, 1000 draws per m. (a) Probability that the repaired threshold controls scene-averaged miscoverage at α (dotted line: $1 - \delta$), AutoBot only. (b) Median region area relative to the oracle, all five source models; hollow markers: threshold infinite in $> 5\%$ of draws.

The direction replicates everywhere: same-domain gap 0.1 points; AV2→nuScenes gap +6.4 points (CI [5.0, 7.9]); AV2→Waymo gap +25.0 points (CI [23.5, 26.9]). Reweighting again makes the nuScenes gap worse (+2.4 points). The audit again has power 1.00 at a 3% false-alarm rate. The loss again concentrates in turns (right-turn drop 20 points vs. 5.5 for straight) and in Singapore (14 points more than Boston), with a similar scene-cluster design effect (3.0 vs. AutoBot's 3.9). The scene-level repair is valid at every m on both Wayformer transfers (Fig. 4b) and reaches the same operating point, $m = 150$ scenes for AV2→nuScenes (area 1.8×) and $m = 150$ for AV2→Waymo (area 1.6×), as the AutoBot models.

The input-level explanation does not replicate in size. Injecting the 2 Hz history into AV2 test inputs costs Wayformer 8.2 points, 129% of its nuScenes gap, and with lane dropout 10.6 points (166%): more than the whole gap, where it was about a quarter for AutoBot. Read this with care. Wayformer reads velocity, acceleration and heading channels that AutoBot ignores, and for an injected trajectory we reconstruct them by finite difference, which gives a piecewise-constant velocity inside each 0.2 s segment that real 2 Hz sensing would not have and may overstate the effect. Lane dropout, which needs no reconstruction, still accounts for 16%.

Magnitude does not replicate. Wayformer's AV2→nuScenes gap is under half of AutoBot's confirmatory 15.1 points, while its AV2→Waymo gap (+25.0) is close to AutoBot's (+23.4). This breaks the exploratory zoo's "more accurate model, larger gap" pattern (Sec. V-A): Wayformer's accuracy is comparable to AutoBot's confirmatory model, yet its nuScenes gap is far smaller. We also find a second, unregistered difference worth reporting rather than hiding. On the *same* nuScenes target, agent-level certify-or-recalibrate was almost useless for AutoBot (56% valid at $m = 40$, 4% at $m = 300$) but mostly valid for Wayformer (82% at $m = 40$, 96% at $m = 300$), on identically structured scenes (same mean 66 agents/scene). The difference tracks the size of the underlying miscalibration: the source threshold misses 26.8% of AutoBot's evaluation

agents against only 17.5% of Wayformer's, so the same scene clustering has more room to hurt a method that was already more wrong. Scene size is therefore not a sufficient predictor of agent-level failure on its own, which is itself a reason to prefer the scene-level guarantee always, rather than judge case by case whether agent-level recalibration happened to get lucky.

G. Exploratory models

Three reduced-budget AV2 models and one reduced-budget nuScenes model (Table I) were analysed before the confirmatory models existed. Each reproduces the sign of every confirmatory finding: under-coverage on the target, no help from reweighting, losses concentrated in turns, and agent-level failure under scene clustering. The magnitudes vary by a factor of three, which is why the claims above rest on the confirmatory models alone.

VI. Discussion

A planner that trusts a source-calibrated region on a new dataset trusts a region that is too small. At $\alpha = 0.1$ the confirmatory models lose 15.1, 7.7, 23.4 and 18.3 points of coverage on the four transfers we tested, and the loss is silent: nothing observable without labels signals it. Four results bear directly on practice.

Label-free fixes are not safe defaults. Covariate reweighting made the gap larger on every forward model, and the normalised score helped in one direction and hurt in the other. A practitioner who applies either and stops can be worse off than one who does nothing.

Auditing is cheap; repairing is not. About 10^3 labelled target agents suffice to detect a gap of this size with high power. Repairing it with a guarantee that holds under scene clustering needs whole labelled scenes: roughly 150 to 200 of them before the region is both finite and within twice the oracle area. Below that budget the correct behaviour is to abstain, which the scene-level rule does.

A learned predictor of scene loss does not replace labels. A classifier trained on source labels to predict whether an agent is missed explained at most 16% of the variance of the true per-scene miss fraction on any target, and also little on held-out source scenes, so it cannot stand in for target labels, for example as a control variate that shortens the label budget.

The unit of labelling matters, and scene size alone does not predict the failure. On nuScenes (mean 66 agents per scene), agent-level guarantees were nearly useless for AutoBot (56% valid at $m = 40$) but mostly valid for Wayformer on the identical scenes (82%, Sec. V-F), tracking how badly each model's source threshold was miscalibrated there (26.8% vs. 17.5% missed), not the scene structure, which was the same for both. Agent-level guarantees did hold on Waymo and AV2, where scenes are small, for every model we tried. A practitioner cannot know in advance which case they are in without the audit of Sec. V-E, which is the practical argument for using the scene-level guarantee by default rather than hoping agent-level recalibration happens to be close enough.

a) Mechanism: Two observations narrow the cause. For AutoBot the measured input differences explain about a quarter of the gap, and the loss concentrates in turns under left-hand traffic, which AV2 does not contain; for that model both point to a change in the conditional behaviour of road users rather than in the scenes themselves, consistent with Proposition 1. The turn pattern replicates on Wayformer, but the input-level explanation does not: its injected input differences reproduce the whole, smaller gap (Sec. V-F), so how much of the loss is protocol and how much is behaviour depends on the model. The exploratory AutoBot zoo showed the most accurate model losing the most coverage; Wayformer breaks this pattern, with accuracy comparable to the confirmatory AutoBot model but under half its nuScenes gap. Accuracy is therefore not a general predictor of gap size across architectures, only a pattern observed within one architecture family trained several ways.

b) Limitations: (i) Two architectures, one trained in both directions; Wayformer confirms the direction, reweighting, turn and repair findings but was only trained forward, so its reverse direction is untested, and its injection result rests on reconstructed kinematics. (ii) For AutoBot the measured input differences leave most of the gap unexplained, and unmeasured causes cannot be ruled out. (iii) Coverage is marginal over agents or scenes, not conditional on a scenario; the manoeuvre breakdown is a diagnostic, not a guarantee. (iv) Proposition 1 covers scale shifts only. (v) The repair budget is measured on three datasets and may differ on others; the registered evaluation design (disjoint pools, exact evaluation) should be reused rather than the numbers. (vi) Seeds: a second training seed of the confirmatory AutoBot model gave an AV2→nuScenes loss of 20.6 points (95% interval 19–22) against 15.1 for the first, meeting our registered criterion (same sign, within a factor of two) but showing that seed-to-seed variation exceeds the sampling interval of a single run. A second Wayformer seed (accuracy 0.941, same gate) gave 7.7 points on nuScenes against 6.4 and 24.2 against 25.0 on Waymo, so its smaller nuScenes gap relative to AutoBot is not a seed artefact. The reverse direction and the injection and recalibration analyses were not re-seeded.

VII. Conclusion

A conformal guarantee for trajectory prediction does not transfer across driving datasets. On three datasets, two architectures and six source-target pairs, a forecaster calibrated at 90% covered between 65 and 84% of target futures, every time under-covering, never over. Reweighting does not repair the loss and no label-free method can in general. Within one architecture, more accurate models lost more coverage; across architectures this did not hold, so accuracy is not a general predictor of how badly the guarantee fails. Labelled target data can repair it: a scene-level Learn-then-Test rule with a correctly capped betting p-value restores the guarantee at every budget we tested, on both architectures, and gives a usable region from about 150 labelled scenes. We recommend that any transferred conformal region be audited on target labels and, where it fails, recalibrated at the scene level before it is used for planning.

Data and Code Availability

Argoverse 2, nuScenes and Waymo Open Motion are third-party datasets available under their own licences from https://www.argoverse.org, https://www.nuscenes.org and https://waymo.com/open. Code, configuration files, the registered hypotheses, the experiment log and all aggregate result files (no raw third-party data) are at https://github.com/hwaqas160/conformal-coverage-transfer and archived under doi:10.5281/zenodo.23054413 (CC BY 4.0). Every reported result, table and figure is regenerated from the result files by the `paper/make_results*.py` scripts. Descriptions of the datasets themselves (the size of the AV2 training set, the 30 of 150 Waymo validation shards used, the 20 s scene length) are stated from the dataset documentation and our conversion logs, not generated.

Conflict of Interest

The authors declare no competing financial interests or personal relationships that could have influenced this work.

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The corresponding author used Anthropic Claude to edit language, draft and revise text, write and debug analysis code, run and monitor experiments, and generate figures and tables from computed results. The authors reviewed and edited all content and take full responsibility for it.

References

[1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[2] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[3] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, "Safe planning in dynamic environments using conformal prediction," *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 5116–5123, 2023.

[4] S. H. Sun and R. Yu, "Copula conformal prediction for multi-step time series prediction," in *International Conference on Learning Representations (ICLR)*, 2024.

[5] H. Huang, S. He, and F. Miao, "CUQDS: Conformal uncertainty quantification under distribution shift for trajectory prediction," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[6] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, and J. V. Deshmukh, "Formal verification and control with conformal prediction," arXiv:2409.00536, 2024.

[7] L. Feng, M. Bahari, K. M. B. Amor, É. Zablocki, M. Cord, and A. Alahi, "UniTraj: A unified framework for scalable vehicle trajectory prediction," in *European Conference on Computer Vision (ECCV)*, 2024.

[8] T. Demmler, J. Häringer, A. Tamke, T. Dang, A. Hegai, and L. Mikelsons, "Beyond features: How dataset design influences multi-agent trajectory prediction performance," arXiv:2507.05098, 2025.

[9] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[10] R. J. Tibshirani, R. F. Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[11] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *The Annals of Statistics*, vol. 51, no. 2, pp. 816–845, 2023.

[12] L. Aolaritei, Z. O. Wang, J. Zhu, M. I. Jordan, and Y. Marzouk, "Conformal prediction under Lévy–Prokhorov distribution shifts: Robustness to local and global perturbations," arXiv:2502.14105, 2025.

[13] I. Gibbs and E. J. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[14] A. N. Angelopoulos, E. J. Candès, and R. J. Tibshirani, "Conformal PID control for time series prediction," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[15] Y. Shu, J. Xu, L. Kong, F. Zhang, G. Yin, and C. Sun, "Scenario-aware uncertainty quantification for trajectory prediction with statistical guarantees," arXiv:2512.05682, 2025.

[16] X. Chen, R. Bhadani, and L. Head, "Conformal trajectory prediction with multi-view data integration in cooperative driving," arXiv:2408.00374, 2024.

[17] N. A. Madjid, A. Ahmad, M. Mebrahtu, Y. Babaa, A. Nasser, S. Malik, B. Hassan, N. Werghi, J. Dias, and M. Khonji, "Trajectory prediction for autonomous driving: Progress, limitations, and future directions," arXiv:2503.03262, 2025.

[18] Z. Zhou, H. Hu, X. Chen, Y. Zhang, N. Guan, Y.-H. Li, C. J. Xue, and J. Wang, "Mode-as-sequence: Translating multimodal motion prediction into unified sequential mode modeling," arXiv:2605.24037, 2026.

[19] N. Song, J. Jiang, J. Li, X. Zhu, and L. Zhang, "UniMotion: A unified motion framework for simulation, prediction and planning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[20] F. C. Pouria, M. Golchoubian, and K. Driggs-Campbell, "Rethinking Gaussian trajectory predictors: Calibrated uncertainty for safe planning," arXiv:2603.10407, 2026.

[21] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, 2018.

[22] A. Doula, M. Mühlhäuser, and A. S. Guinea, "SafePath: Conformal prediction for safe LLM-based autonomous navigation," arXiv:2505.09427, 2025.

[23] Z. Mei, A. Dixit, M. Booker, E. Zhou, M. Storey-Matsutani, A. Z. Ren, O. Shorinwa, and A. Majumdar, "Perceive with confidence: Statistical safety assurances for navigation with learning-based perception," arXiv:2403.08185, 2024.

[24] J. Yao, X. Zhang, Y. Xia, Z. Wang, A. K. Roy-Chowdhury, and J. Li, "Towards generalizable safety in crowd navigation via conformal uncertainty handling," in *Conference on Robot Learning (CoRL)*, 2025.

[25] A. O. Feldman, J. A. Vincent, M. Adang, J. Low, and M. Schwager, "Learning robot safety from sparse human feedback using conformal prediction," arXiv:2501.04823, 2025.

[26] A. E. Binny, M. Anand, H. T. M. Kussaba, L. Chen, S. Agrawal, F. J. Abu-Dakka, and A. Swikir, "Safe and stable neural network dynamical systems for robot motion planning," *IEEE Robotics and Automation Letters*, 2026.

[27] O. Mirzaeedodangeh, E. Shekhtman, N. Matni, and L. Lindemann, "Safe planning in interactive environments via iterative policy updates and adversarially robust conformal prediction," arXiv:2511.10586, 2025.

[28] A. E. Binny and A. Dixit, "Who moved my distribution? Conformal prediction for interactive multi-agent systems," arXiv:2511.11567, 2025.

[29] R. Xu, C. Chen, Y. Sun, P. Venkitasubramaniam, and S. Xie, "Wasserstein-regularized conformal prediction under general distribution shift," arXiv:2501.13430, 2025.

[30] A. H. C. Correia and C. Louizos, "Non-exchangeable conformal prediction with optimal transport: Tackling distribution shifts with unlabeled data," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[31] M. Cauchois, S. Gupta, A. Ali, and J. C. Duchi, "Robust validation: Confident predictions even when distributions shift," *Journal of the American Statistical Association*, vol. 119, no. 548, pp. 3033–3044, 2024.

[32] J. Liu, D. Qiao, Y. Zhang, and X. Chen, "Certify or refuse: A cross-model map for selective risk control with coverage floors under covariate shift," arXiv:2608.10893, 2026.

[33] R. Tumu, A. Singh, and R. Mangharam, "AdaptNC: Adaptive non-conformity scores for conformal prediction under distribution shift," arXiv:2602.01629, 2026.

[34] J. Jun and I. Ohn, "Online conformal inference with retrospective adjustment for faster adaptation to distribution shift," *Pattern Recognition*, vol. 180, 2026.

[35] T. Liu, E. Dobriban, and F. Orabona, "Online conformal prediction via universal portfolio algorithms," arXiv:2602.03168, 2026.

[36] J. Liang, Z. Ren, and Y. Chen, "Optimal training-conditional regret for online conformal prediction," arXiv:2602.16537, 2026.

[37] K. Rahaman, J. V. Deshmukh, A. R. Hota, and L. Lindemann, "When environments shift: Safe planning with generative priors and robust conformal prediction," arXiv:2602.12616, 2026.
[38] R. Dunn, L. Wasserman, and A. Ramdas, "Distribution-free prediction sets for two-layer hierarchical models," *Journal of the American Statistical Association*, vol. 118, no. 544, pp. 2491–2502, 2023.
[39] Y. Lee, R. F. Barber, and R. Willett, "Distribution-free inference with hierarchical data," arXiv:2306.06342, 2023.
[40] G. Principato, G. Stoltz, Y. Amara-Ouali, Y. Goude, B. Hamrouche, and J.-M. Poggi, "Conformal prediction for hierarchical data," arXiv:2411.13479, 2024.
[41] A. Bhattacharyya and R. F. Barber, "Group-weighted conformal prediction," arXiv:2401.17452, 2024.
[42] T. Mortier, A. Javanmardi, Y. Sale, E. Hüllermeier, and W. Waegeman, "Conformal prediction in hierarchical classification with constrained representation complexity," arXiv:2501.19038, 2025.
[43] S. Mallick, E. T. Tchetgen, E. Dobriban, and Y. Lee, "Generalized hierarchical conformal prediction," arXiv:2608.15500, 2026.
[44] Y. Min, L. Peng, and C. Zou, "A unified theory of conditional coverage in conformal prediction with applications," arXiv:2605.11602, 2026.
[45] Z. Zhou, X. Zhang, C. Tao, and Y. Yang, "Conformal prediction assessment: A framework for conditional coverage evaluation and selection," arXiv:2603.27189, 2026.
[46] S. Braun, D. Holzmüller, M. I. Jordan, and F. Bach, "Conditional coverage diagnostics for conformal prediction," arXiv:2512.11779, 2025.
[47] K. Bairaktari, J. Wu, and Z. S. Wu, "Kandinsky conformal prediction: Beyond class- and covariate-conditional coverage," arXiv:2502.17264, 2025.
[48] L. Zhong, X. Wang, S. Huang, and Y. Shi, "When is a conformal guarantee fair? auditing silent subgroup under-coverage in Alzheimer's disease longitudinal prediction," arXiv:2608.04254, 2026.
[49] R. Luo and Z. Zhou, "Conditional conformal risk adaptation," arXiv:2504.07611, 2025.
[50] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. Jordan, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, pp. 43:1–43:34, 2021.
[51] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, 2025.
[52] I. Waudby-Smith and A. Ramdas, "Estimating means of bounded random variables by betting," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 86, no. 1, pp. 1–27, 2024.
[53] J. Alvarez, "A distribution-free valid p-value for finite samples of bounded random variables," arXiv:2405.08975, 2024.
[54] D. T. Nguyen, R. Pathak, A. N. Angelopoulos, S. Bates, and M. I. Jordan, "Data-adaptive tradeoffs among multiple risks in distribution-free prediction," arXiv:2403.19605, 2024.
[55] B.-S. Einbinder, L. Ringel, and Y. Romano, "Semi-supervised risk control via prediction-powered inference," arXiv:2412.11174, 2024.
[56] S. Joshi, T. Wang, H. Hassani, and E. Dobriban, "Risk-controlled post-processing of decision policies," arXiv:2605.06479, 2026.
[57] E. Gauthier, F. Bach, and M. I. Jordan, "e-values expand the scope of conformal prediction," arXiv:2503.13050, 2025.
[58] D. Prinster, X. Han, A. Liu, and S. Saria, "WATCH: Adaptive monitoring for AI deployments via weighted-conformal martingales," in *International Conference on Machine Learning (ICML)*, 2025.
[59] V. Vovk, "Randomness, exchangeability, and conformal prediction," arXiv:2501.11689, 2025.
[60] Q. Wu and M. Campbell, "Semantic and feature guided uncertainty quantification of visual localization for autonomous vehicles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
[61] N. Wang, D. Shang, Y. Gong, X. Hu, Z. Song, L. Yang, Y. Huang, X. Wang, and J. Lu, "Collaborative perception datasets for autonomous driving: A review," arXiv:2504.12696, 2025.
[62] L. Kong, X. Xu, Y. Liu, J. Cen, R. Chen, W. Zhang, L. Pan, K. Chen, and Z. Liu, "LargeAD: Large-scale cross-sensor data pretraining for autonomous driving," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
[63] X. Qiu, W. Liu, F. Ai, Y. Tan, Z. Xu, and C. Song, "Understanding cross-sensor feature variations for generalizable 3D perception," arXiv:2606.11573, 2026.
[64] S. K. Manchingal, A. Amaritei, M. Gohad, M. Sultana, J. F. P. Kooij, F. Cuzzolin, and A. Bradley, "Uncertainty-aware autonomous vehicles: Predicting the road ahead," arXiv:2510.22680, 2025.
[65] V. Vovk, "Conditional validity of inductive conformal predictors," in *Proceedings of the Asian Conference on Machine Learning (ACML)*, ser. Proceedings of Machine Learning Research, vol. 25, 2012, pp. 475–490.
[66] M. Bian and R. F. Barber, "Training-conditional coverage for distribution-free predictive inference," *Electronic Journal of Statistics*, vol. 17, no. 2, pp. 2044–2066, 2023.
[67] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.
[68] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal, B. Pan, R. Kumar, A. Hartnett, J. K. Pontes, D. Ramanan, P. Carr, and J. Hays, "Argoverse 2: Next generation datasets for self-driving perception and forecasting," in *NeurIPS Datasets and Benchmarks Track*, 2021.
[69] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A multimodal dataset for autonomous driving," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
[70] S. Ettinger, S. Cheng, B. Caine, C. Liu, H. Zhao, S. Pradhan, Y. Chai, B. Sapp, C. R. Qi, Y. Zhou, Z. Yang, A. Chouard, P. Sun, J. Ngiam, V. Vasudevan, A. McCauley, J. Shlens, and D. Anguelov, "Large scale interactive motion forecasting for autonomous driving: The Waymo open motion dataset," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
[71] Q. Li, Z. Peng, L. Feng, Z. Liu, C. Duan, W. Mo, and B. Zhou, "ScenarioNet: Open-source platform for large-scale traffic scenario simulation and modeling," in *NeurIPS Datasets and Benchmarks Track*, 2023.
[72] R. Girgis, F. Golemo, F. Codevilla, M. Weiss, J. A. D'Souza, S. E. Kahou, F. Heide, and C. Pal, "Latent variable sequential set transformers for joint multi-agent motion prediction," in *International Conference on Learning Representations (ICLR)*, 2022.
[73] N. Nayakanti, R. Al-Rfou, A. Zhou, K. Goel, K. S. Refaat, and B. Sapp, "Wayformer: Motion forecasting via simple & efficient attention networks," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 2980–2987.



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